

RESEARCH GAP IDENTIFICATION

Project Title

**Explainable Multi-Objective Concrete Mix
Optimization for Strength, Cost, and
Sustainability Using Machine Learning**

Prepared By

MIDUNA R (23CSR131)

NESHA RK (23CSR147)

ROHITH OR (23CSR180)

Department of Computer Science and Engineering
Academic Year: 2023–2027

RESEARCH GAP IDENTIFICATION

Based on the literature survey, the following research gaps have been identified:

1. Lack of Explainability

Most existing machine learning models focus only on achieving high prediction accuracy. However, they do not explain how individual input parameters (such as cement, water, fly ash, and aggregates) influence the predicted compressive strength. This reduces the transparency and trustworthiness of the models.

2. Limited Dataset Size

Many studies use relatively small datasets, which may reduce the generalization capability of machine learning models.

3. Focus on Prediction Rather Than Optimization

Most research work concentrates only on predicting concrete compressive strength. Very few studies optimize the concrete mix proportions to achieve the desired strength while minimizing material usage, cost, and environmental impact.

4. Insufficient Integration of Explainable AI (XAI)

Although Explainable Artificial Intelligence (XAI) techniques such as SHAP are used in some recent studies, their adoption is still limited across existing concrete strength prediction models. More comprehensive integration of explainability is needed to improve model transparency and user confidence.

5. Absence of Concrete Mix Recommendation System

Most existing models predict concrete compressive strength but do not recommend optimized concrete mix proportions based on the desired strength and sustainability requirements.

6. Limited Comparison of Advanced Machine Learning Models

Most existing studies evaluate their models using a single dataset. This limits the model's ability to generalize across different concrete mix designs and real-world conditions. Validation using diverse datasets is needed to improve model reliability and robustness.

7. Lack of User-Friendly Decision Support Systems

Most proposed models remain at the research level and are not developed into practical software tools or user-friendly applications. This limits their adoption by civil engineers and construction professionals.

8. Lack of Cost and Carbon Analysis

Existing studies focus primarily on prediction accuracy and do not estimate construction cost or carbon emissions, limiting their usefulness for sustainable construction.

9. Limited Focus on Sustainable Concrete Mix Design

Only a few studies consider sustainable construction materials such as fly ash, blast furnace slag, and recycled aggregates while optimizing concrete mix design. Greater emphasis is needed on environmentally friendly and sustainable concrete production.

10. Need for an Integrated Explainable Optimization Framework

There is a lack of integrated systems that combine **Machine Learning, Concrete Mix Optimization**, and **Explainable Artificial Intelligence** into a single framework. Developing such a system would improve prediction accuracy, optimize material proportions, enhance transparency, and support better engineering decision-making.

Conclusion of Research Gap

From the identified research gaps, it is evident that there is a need to develop an Explainable Concrete Mix Optimization System using Machine Learning that not only predicts concrete compressive strength accurately but also recommends optimized mix proportions, estimates cost and carbon emissions, provides transparent explanations for predictions, and supports sustainable decision-making in construction.